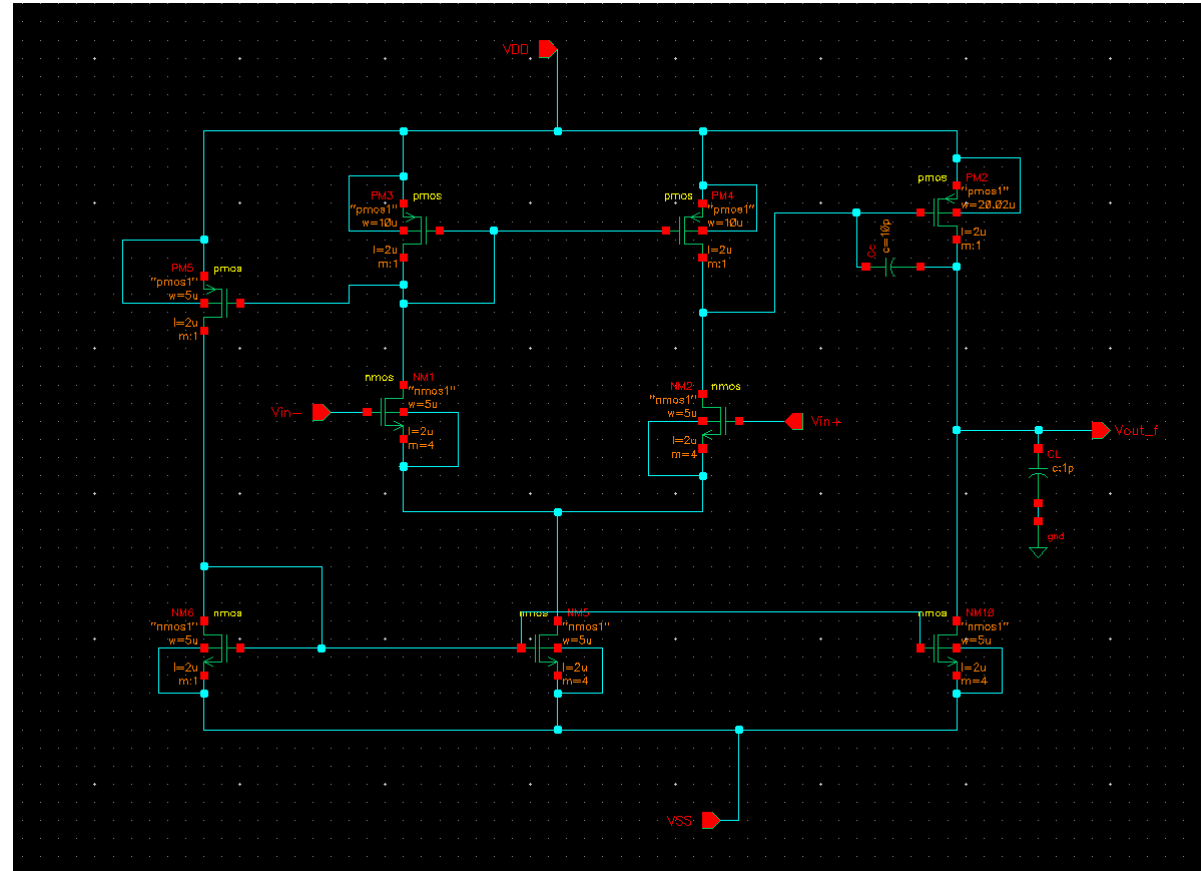


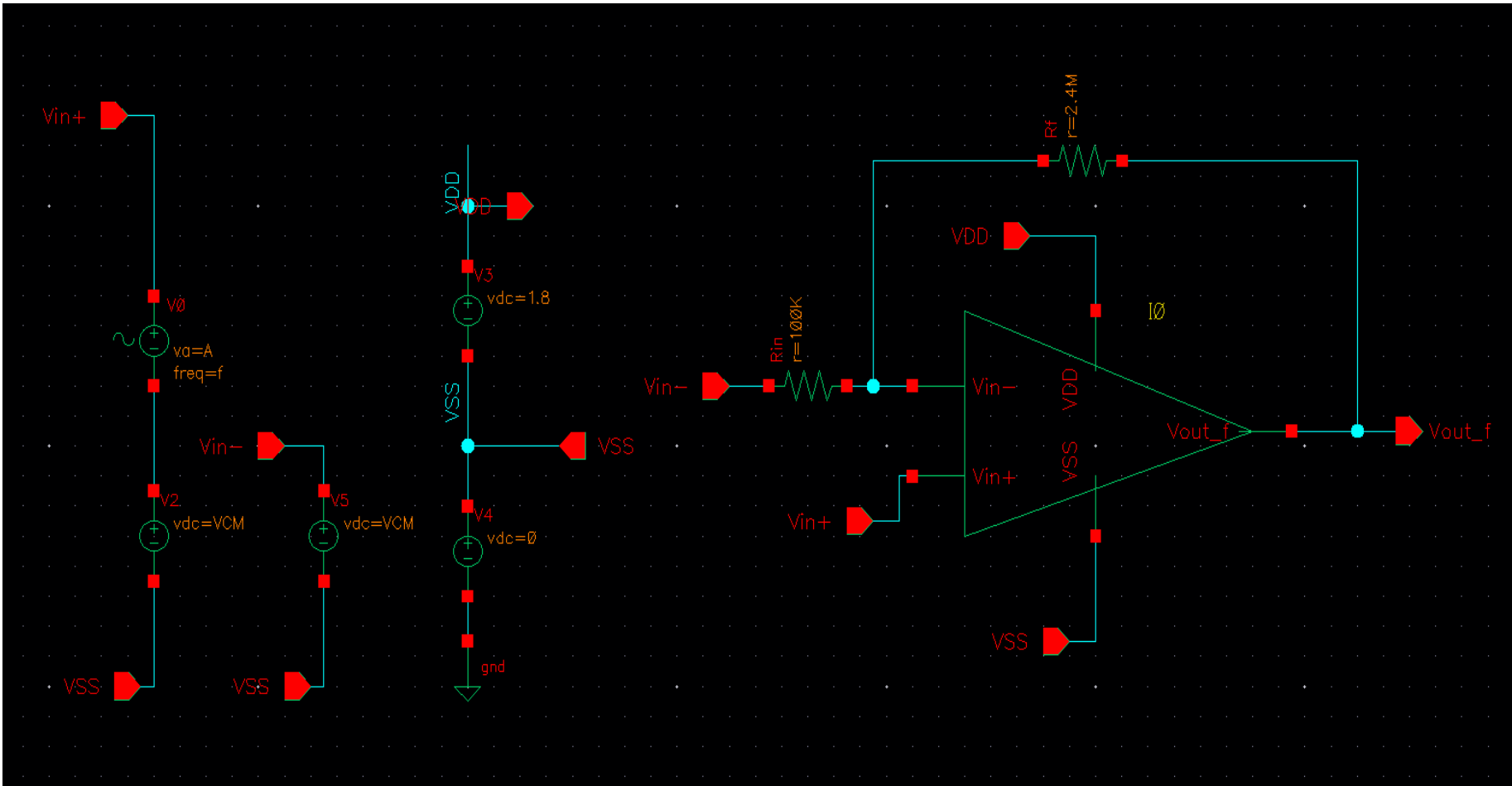
Analog IC Design Project



Submitted By:
Vansh Jain, STCD Batch-2

Non-inverting amplifier with gain of 25 for 10mV sinusoid signal at 10KHz frequency.

Upper Schematic



Gain formula for non-inverting op-amp:

$$A_v = 1 + \frac{R_f}{R_{in}}$$

For Gain = 25,
Taking $R_{in} = 100k\Omega$
We got $R_f = 2.4M\Omega$

Also, I took:

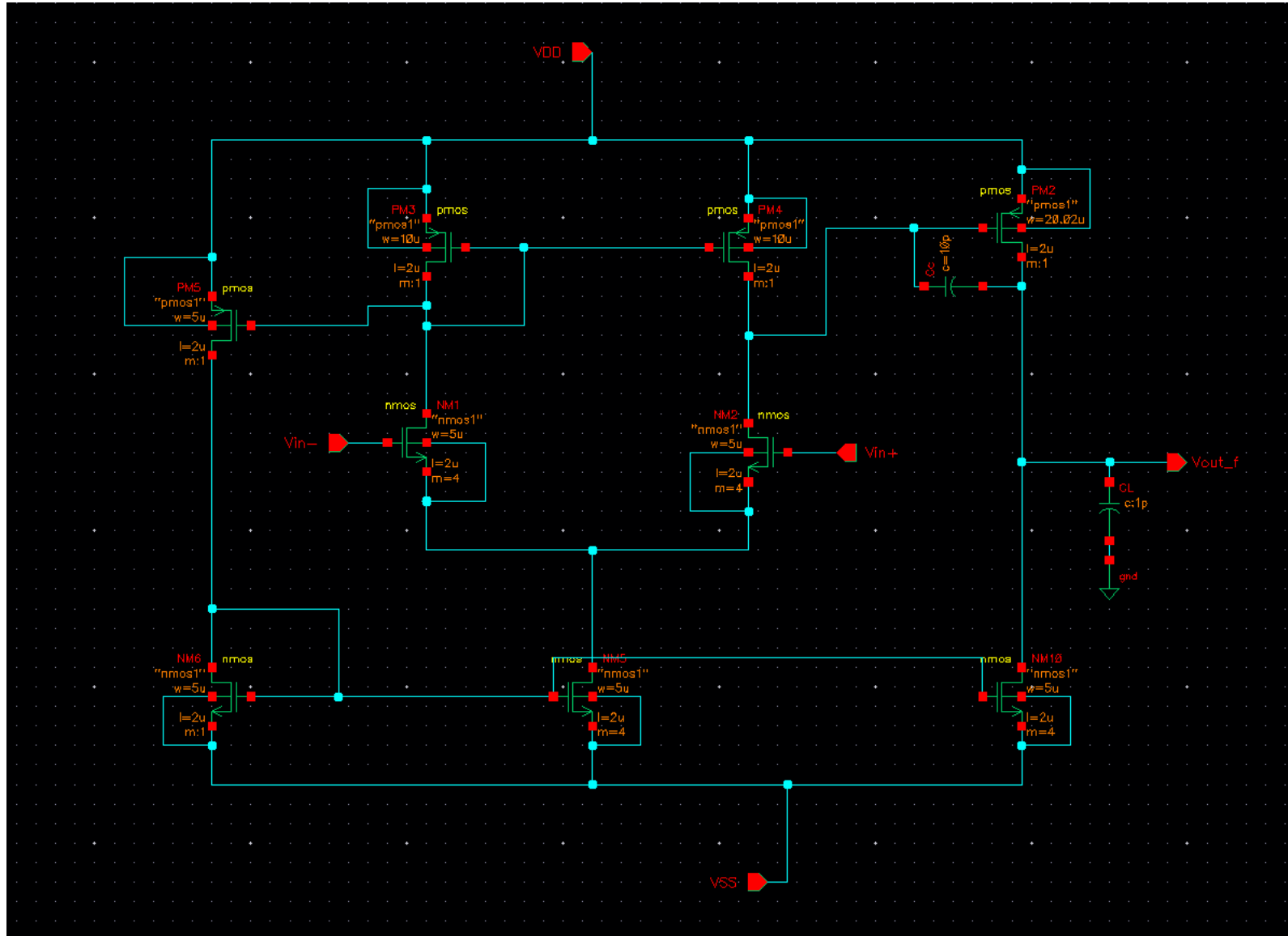
$V_{CM} = 0.9V$

$V_{DD} = 1.8V$

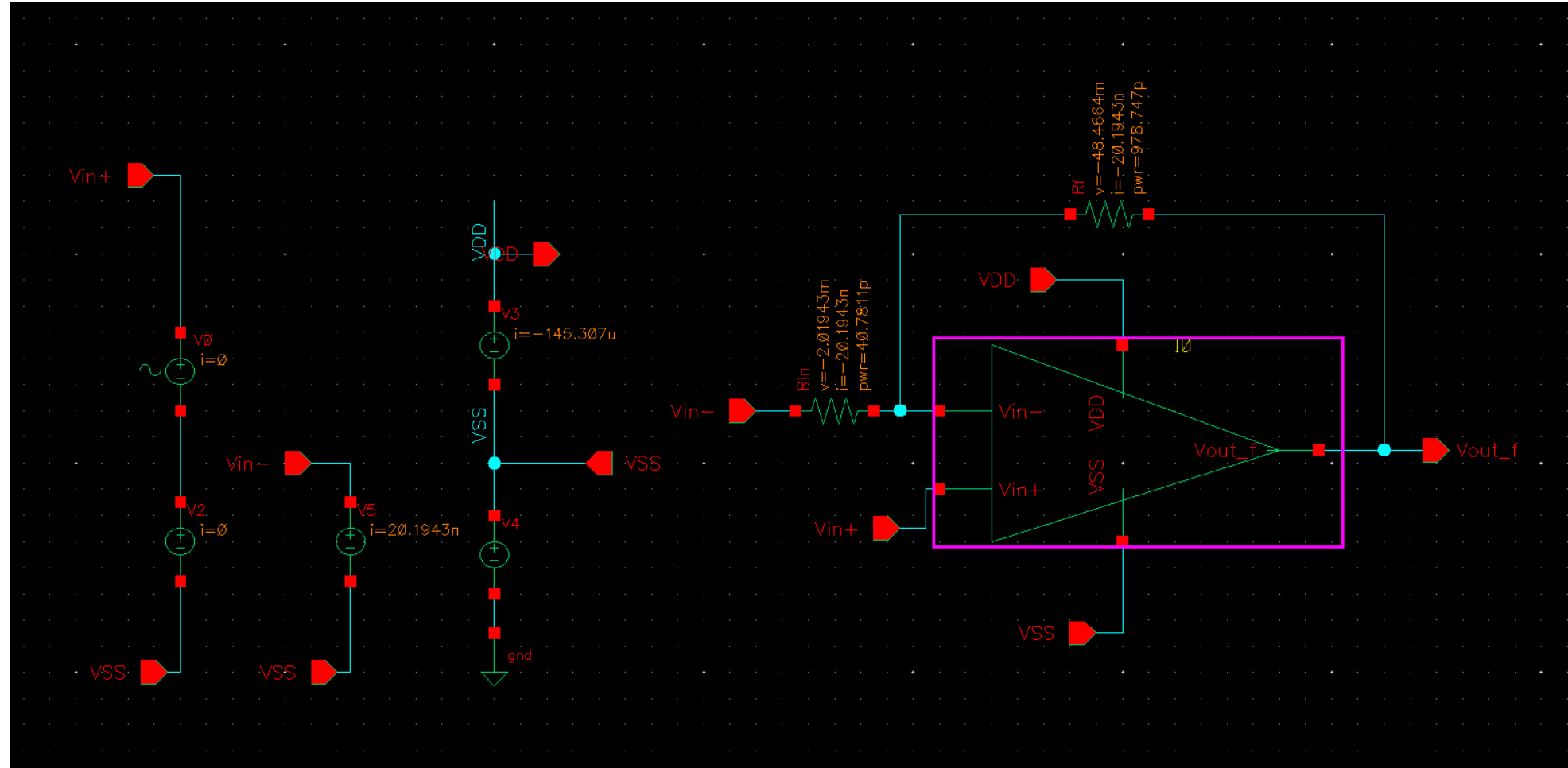
$V_A = 10mA$

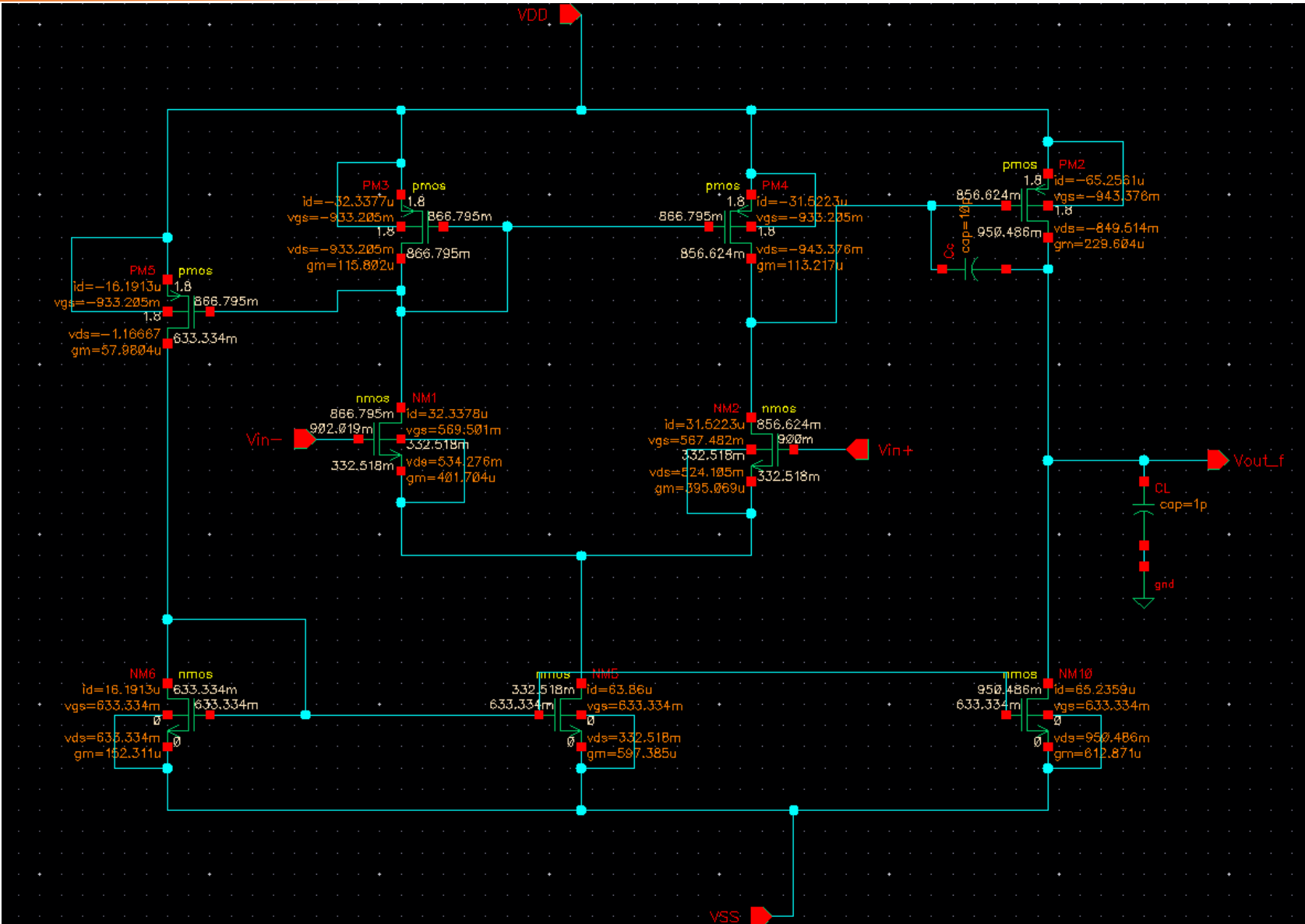
Frequency $f = 10kHz$

Op-Amp Schematic



DC Operating Points





Activities

Log: /home/ITBBS0125_10243/CDS.log

Feb 7 21:14

Virtuoso® ADE Explorer Editing: Analog_IC OpAmp_Simple_tb maestro*

Launch Session Setup Analyses Variables Outputs Simulation Results Tools EAD Parasitics/LDE Window Help

Replace (None)

OpAmp_Simple 10 (OpAmp_Simple) maestro OpAmp_Simple_tb

Filter

Analogue_IC_OpAmp_Simple_tb_1

Analyses

tran 0 1m

dc t

Click to add analysis

Design Variables

A 10m

f 10K

VCM 900m

Click to add variable

Parameters

Corners

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

Explorer Run Summary

0 Corner

Nominal Corner

Outputs Setup

Run Preview

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
expr	pv("I0.NM1" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.NM2" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.NM5" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.NM6" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.NM10" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.PM2" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.PM3" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.PM4" "region" ?result "tranOp")		2	☒		☐	
expr	pv("I0.PM5" "region" ?result "tranOp")		2	☒		☐	

mouse L:

M:

R:

1(4)

No image saved: Possible reasons could be that none of the windows or subwindows are visible or are non-empty.

Results: ExplorerRun.0

Analog_IC OpAmp_Simple_tb schematic

Simulator: spectre aps

Batch

Transient Analysis



Gain formula:

$$A_v = \frac{V_{out_pp}}{V_{in_pp}}$$

Where V_{out_pp} and V_{in_pp} are output and input peak-to-peak voltages respectively.

From Graph:

$$V_{out_pp} = 1.2 - 0.7 = 0.5V$$

$$V_{in_pp} = 20mV \text{ (AC signal)}$$

Therefore, Gain:

$$A_v = \frac{500 mV}{20mV} = 25$$

Activities Import Setup Feb 7 13:27

Virtuoso® ADE Explorer Editing: Analog_IC OpAmp_Simple_tb maestro_2*

Launch Session Setup Analyses Variables Outputs Simulation Results Tools EAD Parasitics/LDE Window Help

Setup OpAmp_Simple Project_L1 OpAmp_Simple_tb maestro_2

Filter Filter

Analogue_IC_OpAmp_Simple_tb_1

Simulator spectre

Analyses

- ☒ tran 0.1m
- ☒ dc t

Click to add analysis

Design Variables

- ☒ A 10m
- ☒ f 10k
- ☒ VCM 0.9

Click to add variable

Parameters

- ☒ Corners
 - ☒ Nominal
 - ☒ Fast Fast
 - ☒ Fast Slow
 - ☒ Neutral
 - ☒ Slow Fast
 - ☒ Slow Slow
- ☐ Reliability Analyses
 - ☐ Monte Carlo Sampling
 - ☐ Checks/Asserts

Click to add corner

Explorer Run Summary

- ☒ 5 Corners
- ☒ Nominal Corner

Outputs Setup Results Run Preview

Detail

Parameter	Nominal	stat	Fast Fast	Fast Slow	Neutral	Slow Fast	Slow Slow
gpdk.scs			FF	FS	NN	SF	SS

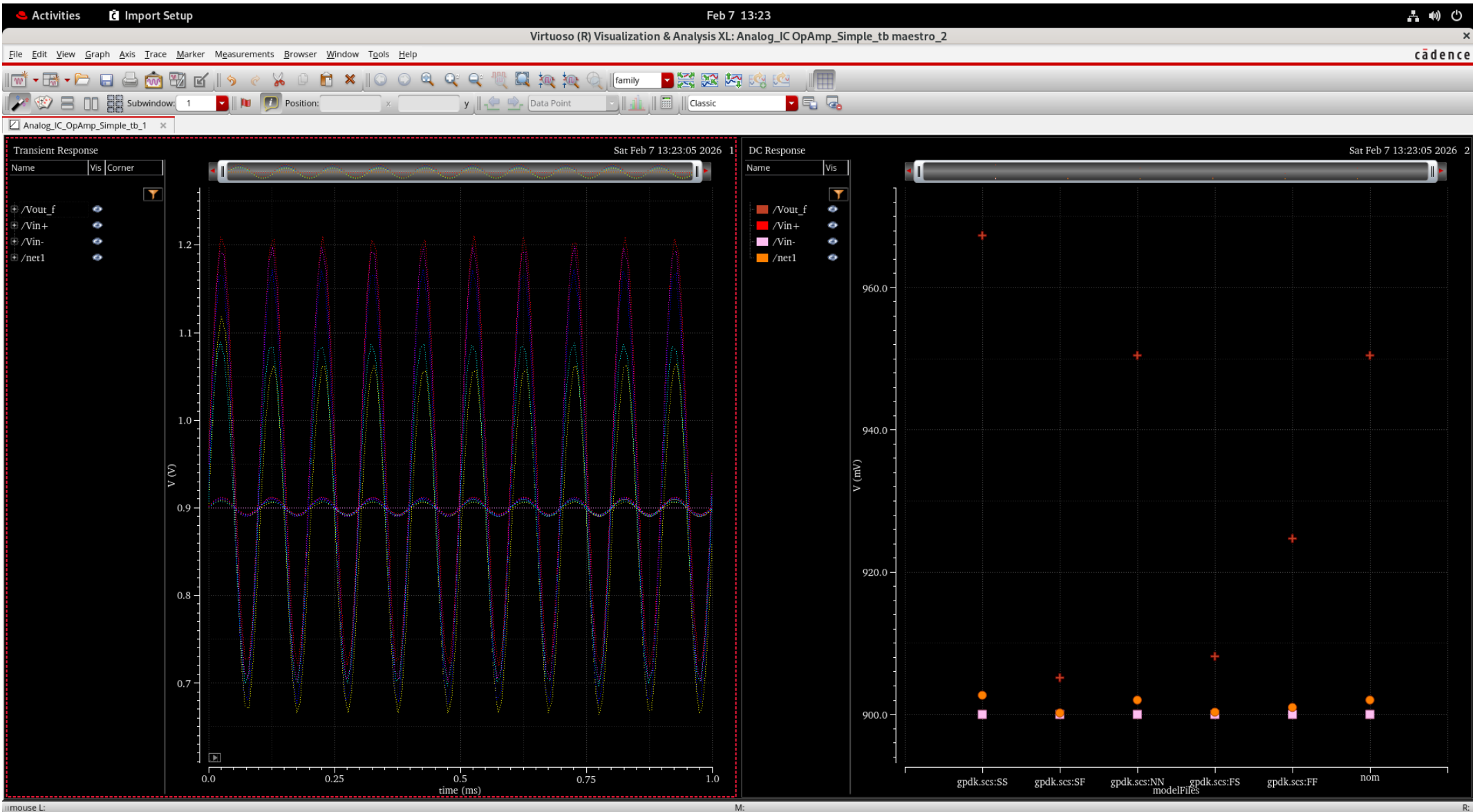
4 rows

Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	Fast Fast	Fast Slow	Neutral	Slow Fast	Slow Slow
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Analogue_IC_OpA...	/vout_f											
Analogue_IC_OpA...	/vin+											
Analogue_IC_OpA...	/vin-											
Analogue_IC_OpA...	/net1											

1(25) | Trace: /vout_f (modelFiles=gpdk.scs:FF); History: ExplorerRun.0; Test: Analogue_IC_OpAmp_Simple_tb_1; Design Point: 1; Corner: Fast Fast

Results: ExplorerRun.0 | Analogue_IC OpAmp_Simple_tb schematic | Simulator: spectre aps

Corner Test (without Temperature setting)



ActivitiesCorners Setup

Feb 7 13:59

Virtuoso® ADE Explorer Editing: Analog_IC OpAmp_Simple_tb maestro_2*

LaunchSessionSetupAnalysesVariablesOutputsSimulationResultsToolsEADParasitics/LDEWindowHelp

Setup

filter

Analog_IC OpAmp_Simple_tb_1

Simulator spectre

Analyses

tran0.1m

dc

Design Variables

A10m

f10k

VCM0.9

Parameters

Corners

Nominal

Fast Fast

Fast Slow

Neutral

Slow Fast

Slow Slow

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

Outputs Setup

Resource CornerNominal

Slow .../Tempera...125

Corners	Nominal	Fast Fast	Fast Slow	Neutral	Slow Fast	Slow Slow
Temperature		-40	0	27	77	125
Design Variables						
Parameters						
Model Files	gpdk.scs	FF	FS	NN	SF	SS
Model Group(s)	<modelgroup>	<modelgroup>	<modelgroup>	<modelgroup>	<modelgroup>	<modelgroup>
Tags						
Number of	1	1	1	1	1	1

OKCancelApplyHelp

Explorer Run Summary

5 Corners

Nominal Corner

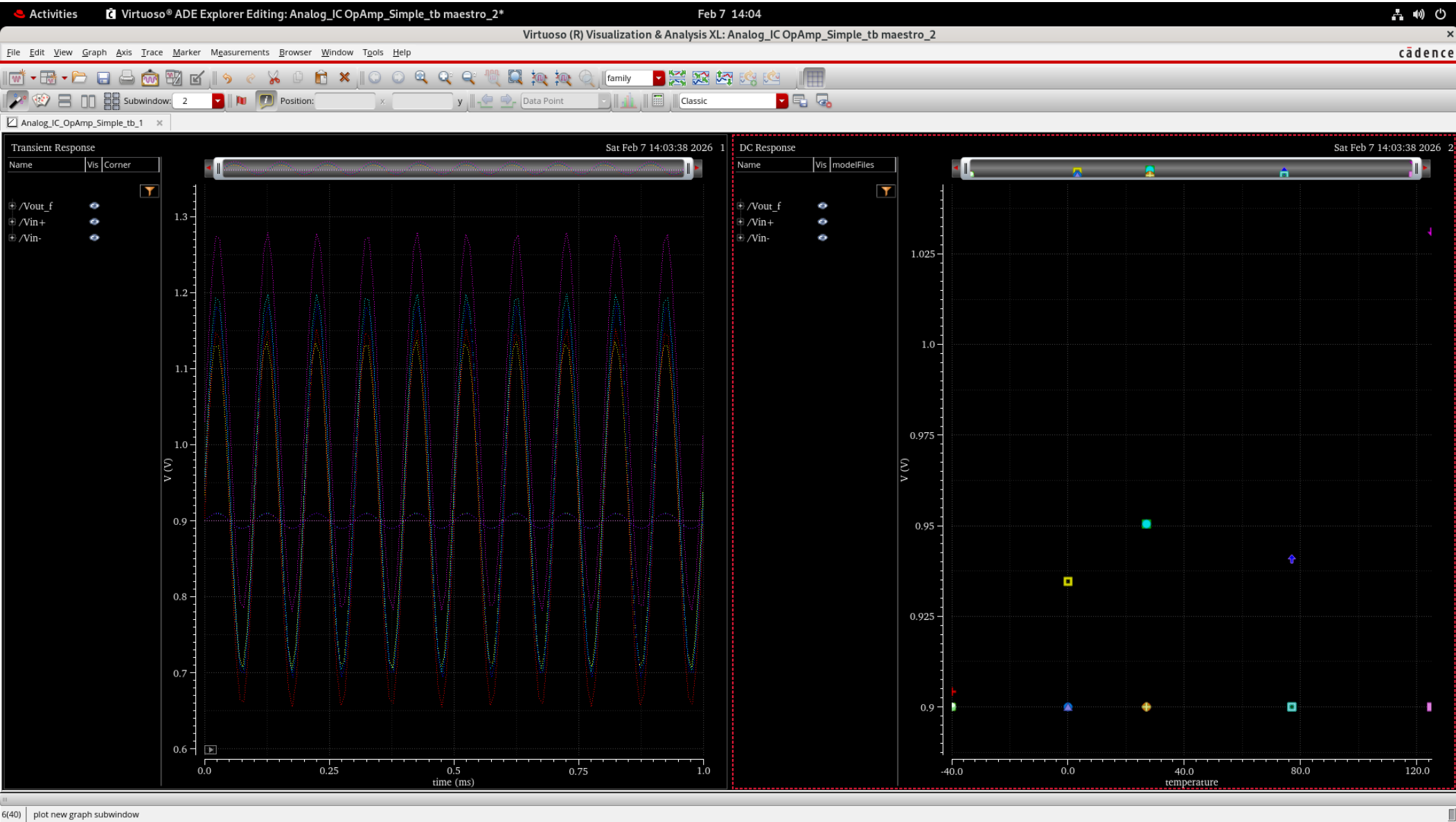
mouse L:5(37)

M:

R:

Analog_IC OpAmp_Simple_tb schematic | Simulator: spectre aps

Corner Test (with Temperature setting)



Monte Carlo Simulation

Activities Import Setup Feb 7 13:30 Virtuoso® ADE Explorer Editing: Analog_IC OpAmp_Simple_tb maestro_2*

Launch Session Setup Analyses Variables Outputs Simulation Results Tools EAD Parasitics/LDE Window Help cadence

Setup

Name	Value
Filter	Filter
Analogue_IC_OpAmp_Simple_tb_1	
Analyses	
tran	0 1m
dc	t
Design Variables	
A	10m
f	10k
VCM	0.9
Parameters	
Corners	
Reliability Analyses	
Monte Carlo Sampling	
Task	Standard Monte Carlo
Number of P...	200
Click to Open Setup Form	
Checks/Asserts	

Explorer Run Summary

5 Corners

Nominal Corner

Monte Carlo

Method: Standard Monte Carlo

Variation: A11

Number of Points: 200

☐ Save Waveforms (Simulation Data)

☐ Save Statistical Parameter Data

Netlist Options

Advanced

Specify Instances/Devices (Not Specified)

Specify Mismatch ID (Not Specified)

Specify Design Variables (Not Specified)

OK Cancel Defaults Help

	Fast Fast	Fast Slow	Neutral	Slow Fast	Slow Slow
FF	FF	FS	NN	SF	SS

Fail	Min	Max	Fast Fast	Fast Slow	Neutral	Slow Fast	Slow Slow
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter

Trace: /Vout_f (modelFiles=gpdK.scs:FF); History: ExplorerRun.0; Test: Analog_IC OpAmp_Simple_tb_1; Design Point: 1; Corner: Fast Fast

Results: ExplorerRun.0 | Analog_IC OpAmp_Simple_tb schematic | Simulator: spectre aps

Monte Carlo Simulation

Activities

Import Setup

Feb 7 13:49

Virtuoso® ADE Explorer Editing: Analog_IC OpAmp_Simple_tb maestro_2*

Launch Session Setup Analyses Variables Outputs Simulation Results Tools EAD Parasitics/LDE Window Help

27

Replace (None)

cadence

OpAmp_Simple Project_L1 OpAmp_Simple_tb maestro_2

Name Value

Analog_IC OpAmp_Simple_tb_1

Simulator: spectre

Analyses

tran 0.1m

dc t

Click to add analysis

Design Variables

A 10m

f 10k

VCM 0.9

Click to add variable

Parameters

Corners

Reliability Analyses

Monte Carlo Sampling

Task Standard Monte Carlo

Number of P... 200

Click to Open Setup Form

Checks/Asserts

Outputs Setup Results Run Preview

Name	Type	Details	Plot	Plot Target	Save	Spec
	expr	average(v("Vout_i") ?result "tran")	☒		☐	

Explorer Run Summary

5 Corners

Nominal Corner

mouse L:

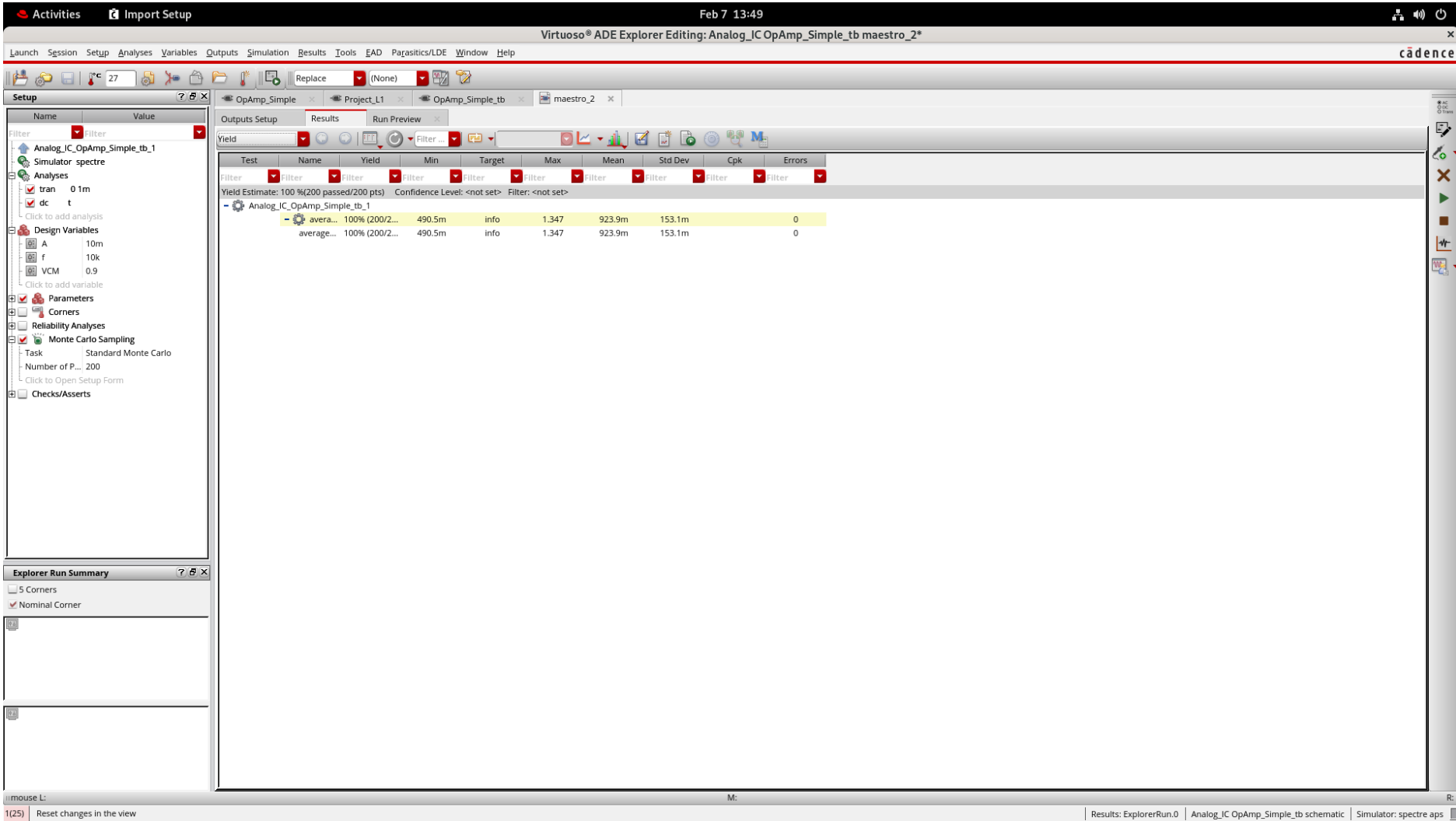
M:

R:

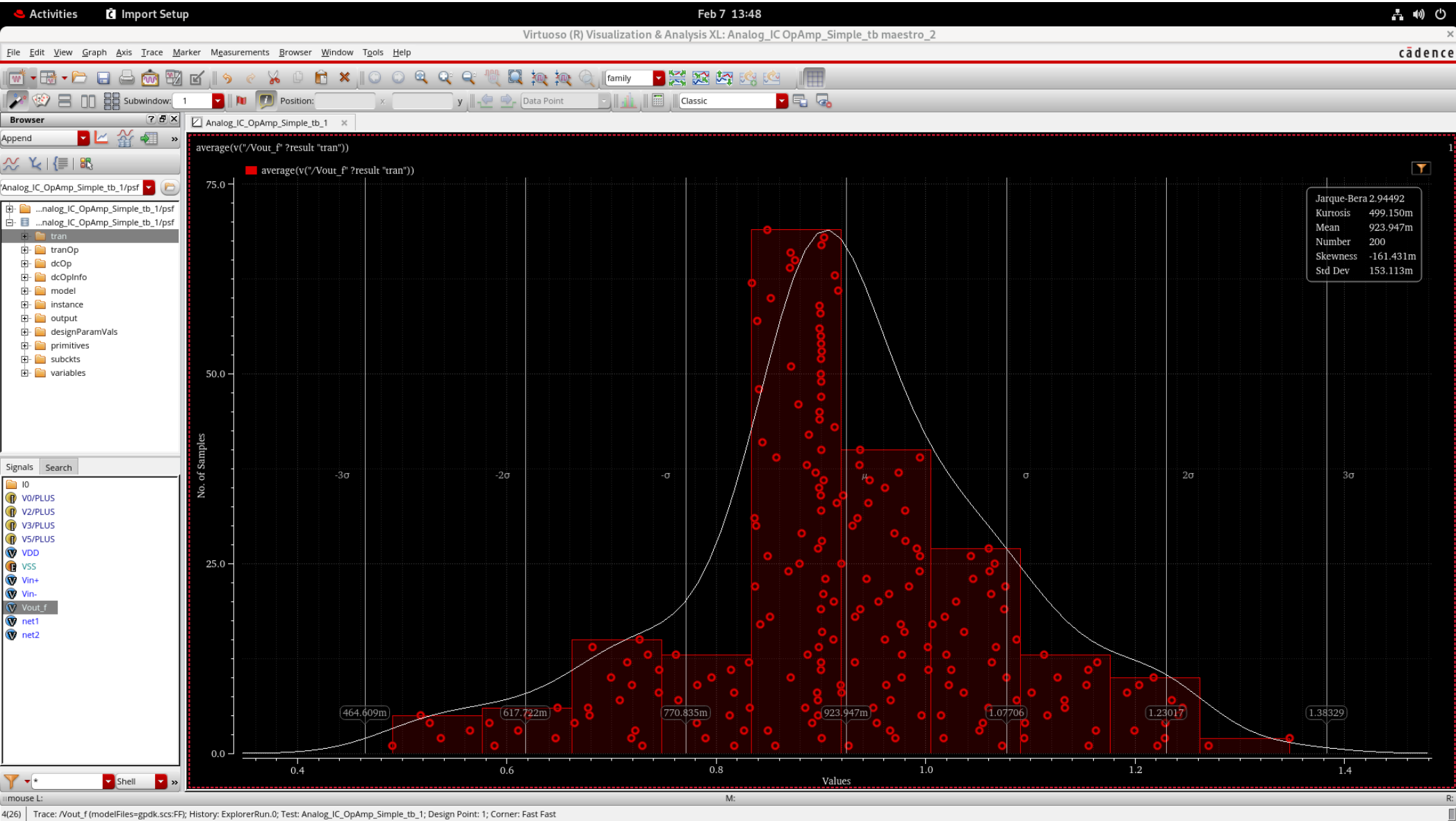
1(25) append plot to current graph

Results: ExplorerRun.0 Analog_IC OpAmp_Simple_tb schematic Simulator: spectre aps

Monte Carlo Simulation



Monte Carlo Simulation



Thank you